Program : **Diploma in Polymer Technology**	
Course Code : **3091**	Course Title: **Natural Rubber Production**
Semester : **3**	Credits: **3**
Course Category: **Program Core**	
Periods per week: **3 (L:2, T:1, P:0)**	Periods per semester: **45**

Course Objectives:

- To provide an idea about different rubber yielding plants, their propagation methods and exposure to various latex extraction methods.
- To give an idea about preservation and concentration of NR latex.
- To get an idea about the dry marketable forms and modified forms of Natural Rubber.

Course Prerequisites:

Topic	Course code	Course name	Semester
Colloids and its properties		Applied Chemistry	1
Monomers, classification of polymers		Fundamentals of Polymer Science	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify different rubber yielding plants, their propagation methods and extraction of NR	11	Understanding
CO2	Explain the composition, preservation and concentration of NR latex	12	Understanding
CO3	Explain the processing of NR latex in to dry marketable forms	10	Understanding
CO4	Describe the production, properties and applications of specialty rubbers and reclaimed rubber	10	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Identify different rubber yielding plants, their propagation methods and extraction of NR**		
M 1.01	Explain the importance and uniqueness of NR	2	Remembering
M 1.02	Describe various propagation methods and classification of clones in terms of rubber plants	3	Understanding
M1.03	Explain tapping and various processes associated with extraction of latex from rubber trees.	3	Understanding
M1.04	Explain method of application and importance of rain guarding and yield stimulants.	3	Understanding
Contents: Natural rubber - uniqueness, structure and properties, major sources statistics of production and consumption - India and World. Rubber yielding plants - propagation methods - brown and green budding - different clones, tapping - standard of tappability - tapping task - tapping rest - tapping system. Types of tapping - Ladder, slaughter and puncture tapping. Tapping systems in India, yield stimulants and rain guarding			
CO2	**Explain the composition, preservation and concentration of NR latex**		
M2.01	Explain the composition and colloid nature of Natural rubber latex	2	Understanding
M2.02	Describe the various short term and long term preservative systems used in NR latex	4	Understanding
M2.03	Explain the major concentration methods of NR latex.	4	Understanding
M2.04	Describe the production of value added products from latex like double centrifuged latex and skim rubber	2	Understanding
	Series Test - I	1	

Contents:

Definition of NR latex - composition and colloid nature - role of micro organisms in pre coagulation - short term and long term preservation systems - Anti coagulants, Concentration of NR latex - creaming, centrifuging, principle process machinery and plant lay out. BIS specification of centrifuged latex. Double centrifuged latex, skim rubber.

CO3	**Explain the processing of NR latex in to dry marketable forms**		
M3.01	List the different marketable forms of dry natural rubber	1	Remembering
M3.02	Describe the machinery and process of manufacture of sheet rubber.	3	Understanding
M3.03	Describe the machinery and process of manufacture of crepe rubber.	3	Understanding
M3.04	Describe the machinery and process of manufacture of technically specified block rubber.	3	Understanding

Contents:

Sheet rubber - Ribbed smoked sheets, Air dried sheets Processing, machinery, Grading - Green book. Crepe rubber - processing, machinery, different grades. ISNR-Processing, and machinery - grades - BIS specification.

CO4	**Describe the production, properties and applications of specialty rubbers and reclaimed rubber**		
M4.01	List the different specialty forms of Natural Rubber.	1	Remembering
M4.02	Explain the production of superior processing rubbers	2	Understanding
M4.03	Explain processing of Natural rubber latex into Constant viscosity and low viscosity rubbers.	2	Understanding
M4.04	Describe the production of OENR and Tyre rubber.	2	Understanding
M4.05	Describe digestor and reclamator process for the production of reclaimed rubber. Properties and application of hawai crumb rubber (buffed rubber	3	Understanding
	Series Test - II	1	

Contents:

Specialty rubbers - production - advantages and disadvantages - superior processing rubbers - PA 50 and PA 80, CV and LV rubbers, OENR, Tyre rubber.

Reclaim rubber - List various production process, Reclamator process, Digester process - Advantages of Reclaim rubber Properties and application of hawai crumb rubber

Text / Reference:

T/R	Book Title/Author
R1	Hand book of natural rubber production -Rubber board
R2	High polymer lattices vol 1 -D C Blackley (Publisher - Springer Netherlands, 1997 Edition)
R3	High Polymer Lattices, vol-I, vol-II, Vol-III - D.C.Blackley (Publisher - SpringerNetherlands,1997)
R4	Latex in Industry - R.J.Noble (Publisher - Rubber Age, Palmerton, New York).
R5	Training Manual - Rubber Training Institute, RRII, Kottayam
R6	Rubbery Materials and their compounds - J.A Brydson(Publisher -Elsevier Applied Science, London, 1988)

Online Resources:

Sl.No	Website Link
1	http://www.rubberscience.in.
2	https://www.researchgate.net
3	https://www.scribd.com
4	http://dl.lib.mrt.ac.lk
\5	http://www.indiannaturalrubber.com